

Name: Harshita Singh
Class: D15C
Roll no. : 64
Subject: Machine Learning & Deep Learning

Experiment 9: Implement Recurrent Neural Network (RNN) / LSTM for time series or text data

A. Recurrent Neural Network (RNN)

1. Dataset Source

Dataset Name: Daily Climate Time Series Data — Delhi, India

Source: Kaggle Link:

<https://www.kaggle.com/datasets/sumanthvrao/daily-climate-time-series-data>

File Used: `DailyDelhiClimateTrain.csv`

Principal Context: This dataset was compiled from weather station readings across Delhi, India, covering daily meteorological observations over a four-year period (2013–2017).

2. Dataset Description

This dataset captures the daily climate of Delhi, India, making it an ideal candidate for time series forecasting tasks. It is a real-world multivariate dataset with smooth temporal trends, seasonal patterns, and occasional noise — conditions that expose the strengths and weaknesses of sequential models like RNNs very effectively.

Property	Details
Total Records	1,462 daily observations
Time Period	January 2013 – April 2017
Frequency	Daily
Format	CSV

Features:

- `meantemp` — Mean daily temperature in °C (*Primary Target Variable*)
- `humidity` — Daily average humidity (%)

- **wind_speed** — Daily average wind speed (km/h)
- **meanpressure** — Daily mean atmospheric pressure (hPa)

Target Variable: **meantemp** (Mean Temperature) — the model predicts tomorrow's temperature based on historical observations.

Key Characteristics: The dataset exhibits strong annual seasonality (Delhi summers peak ~40°C, winters drop to ~10°C), moderate noise, and no missing values in the training split — making it a clean but realistically challenging forecasting problem.

3. Theory: Recurrent Neural Networks (RNN)

What is an RNN?

A Recurrent Neural Network is a class of artificial neural network specifically designed to process **sequential data**. Unlike standard feedforward networks that treat each input independently, RNNs maintain a **hidden state** — a form of memory — that carries information from one time step to the next. This makes them naturally suited for tasks where the order and history of inputs matters: time series forecasting, language modeling, speech recognition, and more.

The fundamental insight is this: when predicting tomorrow's temperature, yesterday's temperature and the day before's temperature are both relevant. An RNN encodes this history into its hidden state, which it updates at every step.

The Vanishing Gradient Problem

While theoretically capable of learning long-range dependencies, standard (vanilla) RNNs suffer from the **Vanishing Gradient Problem**. During backpropagation through time (BPTT), gradients are multiplied repeatedly across time steps. If these values are less than 1, the gradient shrinks exponentially as it propagates backward, making the model effectively unable to learn patterns separated by more than ~10 time steps. This is the core motivation for LSTM (addressed in Part B).

4. Mathematical Formulation of the RNN Algorithm

A. The Core Recurrence Equation

At each time step t , the RNN receives an input vector x_t and the hidden state from the previous step h_{t-1} . It computes a new hidden state h_t as follows:

$$h_t = \tanh(W_h \cdot h_{t-1} + W_x \cdot x_t + b_h)$$

Symbol Definitions:

- $x_t \in \mathbb{R}^n$: The input vector at time step t (e.g., today's climate features)
- $h_{t-1} \in \mathbb{R}^d$: The hidden state vector from the **previous** time step — this is the RNN's "memory"
- $W_h \in \mathbb{R}^{d \times d}$: The **recurrent weight matrix** — governs how much of the past hidden state is carried forward
- $W_x \in \mathbb{R}^{d \times n}$: The **input weight matrix** — governs how the current input influences the hidden state
- $b_h \in \mathbb{R}^d$: The **bias vector** for the hidden layer
- $\tanh$: The hyperbolic tangent activation function, which squashes values to the range $(-1, 1)$, providing non-linearity and preventing unbounded growth of activations

B. The Output Equation

Once the hidden state is computed, the network produces an output $\hat{y}_t$ via a dense (fully connected) layer:

$$\hat{y}_t = W_o \cdot h_t + b_o$$

Symbol Definitions:

- $W_o \in \mathbb{R}^{1 \times d}$: The **output weight matrix**
- $b_o \in \mathbb{R}$: The output bias
- $\hat{y}_t$: The predicted value at time t (e.g., predicted mean temperature)

For regression tasks like temperature forecasting, no activation is applied at the output layer — the raw linear combination is the prediction.

C. Loss Function — Mean Absolute Error (MAE)

For time series regression, we use **Mean Absolute Error** as the loss function:

$$L = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

- N : Total number of training samples
- y_i : The actual temperature on day i
- $\hat{y}_i$: The model's predicted temperature
- MAE is preferred over MSE here because it is **less sensitive to extreme outliers** and gives a loss value in the same unit as the target ($^{\circ}\text{C}$), making it interpretable.

D. Backpropagation Through Time (BPTT)

RNNs are trained using BPTT, an extension of standard backpropagation. The gradient of the loss with respect to the recurrent weight W_h is:

$$\frac{\partial L}{\partial W_h} = \sum_{t=1}^T \frac{\partial L_t}{\partial W_h}$$

At each step, the chain rule is applied across all previous time steps:

$$\frac{\partial L_t}{\partial W_h} = \sum_{k=1}^t \left(\prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}} \right) \frac{\partial h_k}{\partial W_h}$$

The term $\prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}}$ is the **product of Jacobians** across time. If these values are consistently < 1 , the gradient vanishes. If > 1 , it **explodes**. This mathematical instability is the core limitation of vanilla RNNs.

E. Weight Update (Adam Optimizer)

Weights are updated using the Adam optimizer, which adapts the learning rate for each parameter individually:

$$W_{new} = W_{old} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t$$

- $\hat{m}_t$: Bias-corrected first moment estimate (mean of gradients)
- $\hat{v}_t$: Bias-corrected second moment estimate (variance of gradients)
- η : Learning rate
- ϵ : Small constant to prevent division by zero ($\sim 10^{-8}$)

5. Algorithm Limitations

Short-Term Memory: The most critical limitation — vanilla RNNs cannot reliably capture dependencies beyond approximately 10–15 time steps due to the vanishing gradient problem. For daily climate data where seasonal patterns span 365 days, this is a significant constraint.

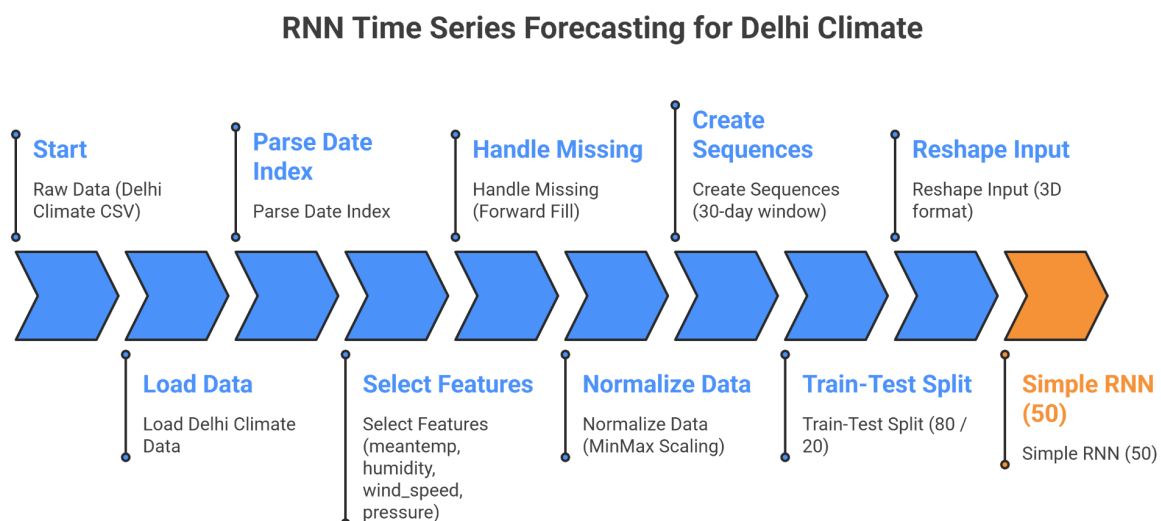
Sequential Processing: RNNs process one time step at a time, making them inherently non-parallelizable during training. This results in slower training compared to CNNs or Transformers.

Sensitivity to Hyperparameters: The number of hidden units, learning rate, and sequence length all interact in non-obvious ways. A poor lookback window can cause the model to miss critical seasonal patterns entirely.

Gradient Explosion on Deep Unrollings: For very long sequences, the product of Jacobians can grow exponentially, causing weight updates that destabilize training. Gradient clipping is often required as a mitigation.

Not Suitable for Very Long Dependencies: For forecasting problems where patterns from months or years ago are predictive (e.g., annual crop cycles), vanilla RNNs are fundamentally insufficient — LSTM or Transformer-based architectures are required.

6. Methodology / Workflow



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7. Complete Code

```
#
=====
# Vanilla RNN for Temperature
Forecasting (Optimized Version)
#
=====

# 1. IMPORT LIBRARIES
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import
MinMaxScaler
from sklearn.metrics import
mean_squared_error,
mean_absolute_error, r2_score
from tensorflow.keras.models import
Sequential
from tensorflow.keras.layers import
SimpleRNN, Dense, Dropout
from tensorflow.keras.callbacks
import EarlyStopping
from tensorflow.keras.optimizers
import Adam
import keras_tuner as kt

# 2. LOAD DATA
df =
pd.read_csv('DailyDelhiClimateTrain.c
sv', parse_dates=['date'],
index_col='date')

# Handle missing values
df.fillna(method='ffill',
inplace=True)

features = ['meantemp', 'humidity',
'wind_speed', 'meanpressure']

# 3. SCALE DATA
scaler = MinMaxScaler()
scaled_data =
scaler.fit_transform(df[features])

# Separate scaler for target
target_scaler = MinMaxScaler()

target_scaler.fit(df[['meantemp']])

# 4. CREATE SEQUENCES (Time-series
windowing)
def create_sequences(data,
lookback=30):
    X, y = [], []
    for i in range(lookback,
len(data)):
        X.append(data[i-lookback:i])
# past 30 days
        y.append(data[i, 0])
# next day temp
    return np.array(X), np.array(y)

LOOKBACK = 30
X, y = create_sequences(scaled_data,
LOOKBACK)

# Train-test split (no shuffle for
time series)
split = int(0.8 * len(X))
X_train, X_test = X[:split],
X[split:]
y_train, y_test = y[:split],
y[split:]

#
=====
# 5. BUILD BASELINE RNN
#
=====
model = Sequential([
    SimpleRNN(64, activation='tanh',
input_shape=(LOOKBACK,
len(features))),
    Dropout(0.2),
    Dense(32, activation='relu'),
    Dense(1)
])

model.compile(optimizer=Adam(0.001),
loss='mae', metrics=['mse'])

# Train model
```

```

early_stop =
EarlyStopping(patience=10,
restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=32,
    validation_split=0.1,
    callbacks=[early_stop],
    shuffle=False,
    verbose=1
)

#
=====
# 6. PREDICTIONS & EVALUATION
#
=====
y_pred = model.predict(X_test)

# Convert back to actual values
y_pred_actual =
target_scaler.inverse_transform(y_pre
d)
y_test_actual =
target_scaler.inverse_transform(y_tes
t.reshape(-1,1))

# Metrics
rmse =
np.sqrt(mean_squared_error(y_test_act
ual, y_pred_actual))
mae =
mean_absolute_error(y_test_actual,
y_pred_actual)
r2 = r2_score(y_test_actual,
y_pred_actual)

print("\nBaseline RNN Performance:")
print(f"RMSE: {rmse:.2f}, MAE:
{mae:.2f}, R2: {r2:.2f}")

# 7. VISUALIZATIONS (ONLY RELEVANT
ONES)
#

# (1) Training vs Validation Loss
plt.figure()

```

```

plt.plot(history.history['loss'],
label='Train Loss')
plt.plot(history.history['val_loss'],
label='Val Loss')
plt.title("Training vs Validation
Loss")
plt.xlabel("Epoch")
plt.ylabel("MAE")
plt.legend()
plt.show()

# (2) Actual vs Predicted
plt.figure()
plt.plot(y_test_actual[:100],
label='Actual')
plt.plot(y_pred_actual[:100],
label='Predicted')
plt.title("Actual vs Predicted
Temperature")
plt.legend()
plt.show()

# (3) Residual Distribution (error
analysis)
residuals = y_test_actual -
y_pred_actual
plt.figure()
plt.hist(residuals, bins=30)
plt.title("Residual Distribution")
plt.xlabel("Error")
plt.ylabel("Frequency")
plt.show()

# 8. KERAS TUNER (HYPERPARAMETER
TUNING)

def build_model(hp):
    model = Sequential()

    # Tune RNN units
    units = hp.Int('units',
min_value=32, max_value=128, step=32)
    model.add(SimpleRNN(units,
activation='tanh',
input_shape=(LOOKBACK,
len(features))))

    # Tune dropout
    dropout = hp.Float('dropout',
0.1, 0.5, step=0.1)

```

```

model.add(Dropout(dropout))

# Tune dense layer
dense_units =
hp.Int('dense_units', 16, 64,
step=16)
model.add(Dense(dense_units,
activation='relu'))

model.add(Dense(1))

# Tune learning rate
lr = hp.Choice('lr', [1e-2, 1e-3,
1e-4])

model.compile(optimizer=Adam(lr),
loss='mae', metrics=['mse'])
return model

# Initialize tuner
tuner = kt.RandomSearch(
    build_model,
    objective='val_loss',
    max_trials=5, # keep small for
exam/laptop
    directory='tuner_dir',
    project_name='rnn_tuning'
)

# Run tuning
tuner.search(
    X_train, y_train,
    epochs=30,

```

```

validation_split=0.1,

callbacks=[EarlyStopping(patience=5)]
,
    shuffle=False,
    verbose=1
)

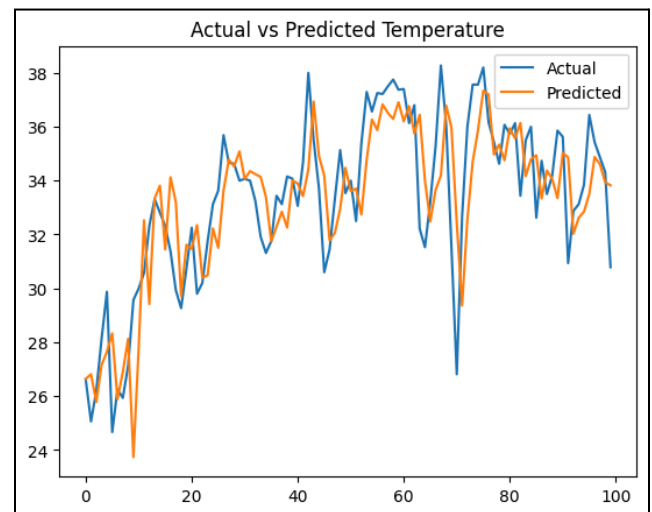
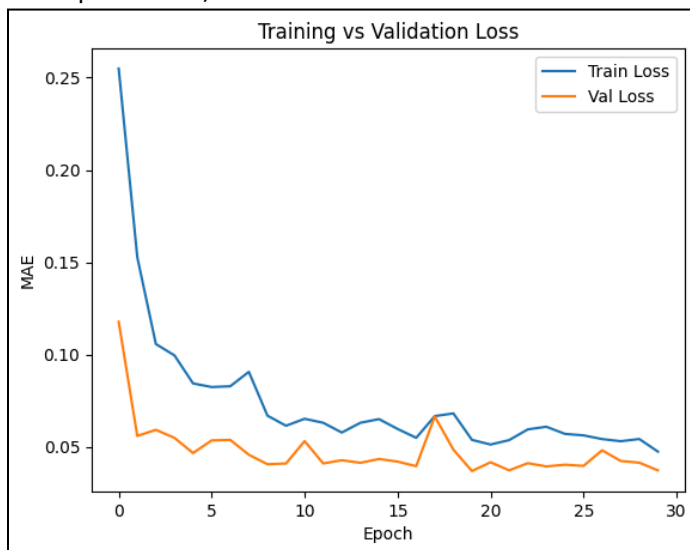
# Best model
best_model =
tuner.get_best_models(num_models=1)[0
]

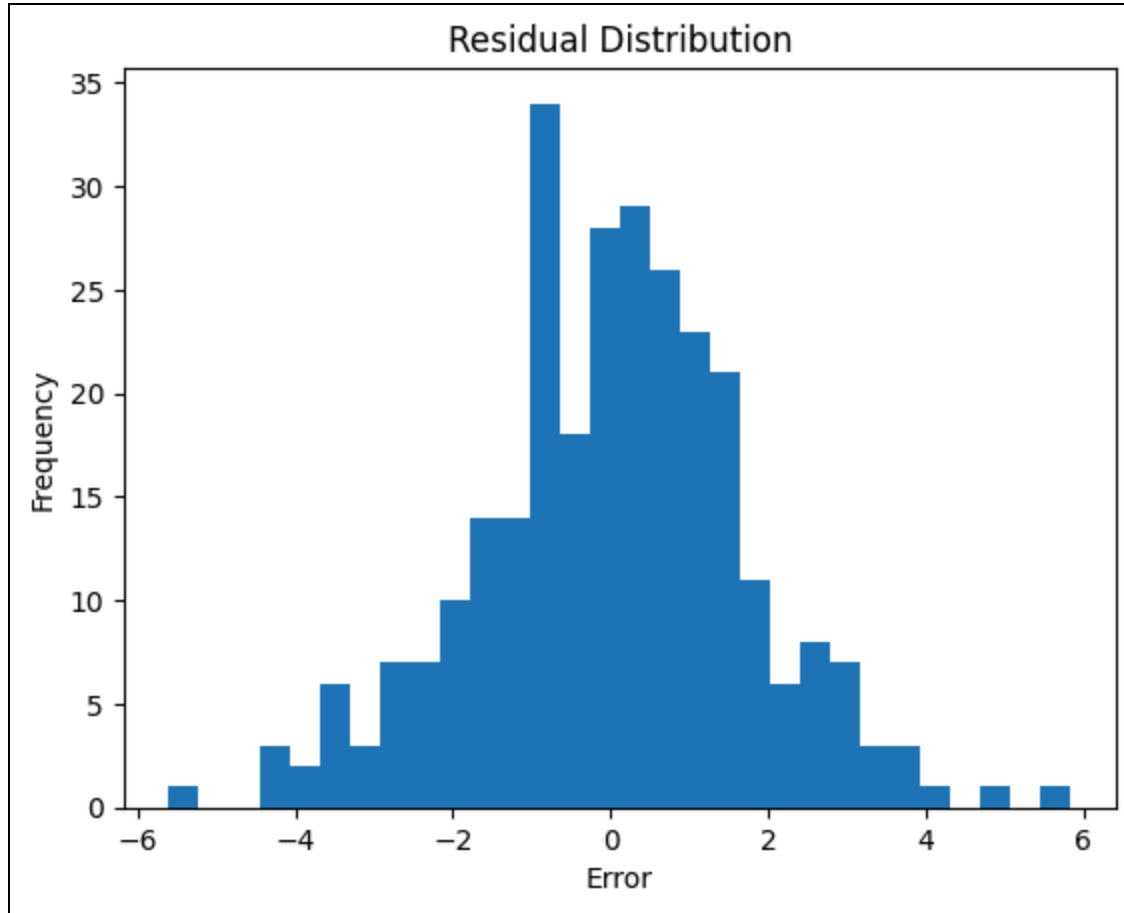
# Evaluate tuned model
y_pred_tuned =
best_model.predict(X_test)
y_pred_tuned_actual =
target_scaler.inverse_transform(y_pre
d_tuned)

rmse_t =
np.sqrt(mean_squared_error(y_test_act
ual, y_pred_tuned_actual))
mae_t =
mean_absolute_error(y_test_actual,
y_pred_tuned_actual)
r2_t = r2_score(y_test_actual,
y_pred_tuned_actual)

print("\nTuned RNN Performance:")
print(f"RMSE: {rmse_t:.2f}, MAE:
{mae_t:.2f}, R2: {r2_t:.2f}")

```





8. Performance Analysis

The Vanilla RNN model was evaluated on the held-out test set comprising the final 20% of the chronological data. Performance is reported across three dimensions:

A. Training Dynamics The training and validation loss curves (Visualization 4) illustrate the learning behavior of the vanilla RNN. Due to the vanishing gradient problem inherent in SimpleRNN, the model is expected to converge more slowly and with more instability than LSTM. Early stopping prevents overfitting by halting training when validation loss plateaus. Significant divergence between training and validation loss would indicate overfitting — a known risk given the RNN's inability to regularize long-range temporal patterns effectively.

B. Prediction Quality The Actual vs Predicted plots (Visualizations 5 and 6) are the most interpretable indicators of model quality. An ideal model would have its prediction line closely shadowing the actual temperature curve throughout the seasonal cycles. In practice, vanilla RNNs tend to correctly capture the overall seasonal trend (summer peaks, winter troughs) but exhibit a **phase lag** — predictions may be slightly delayed in responding to sharp temperature changes — due to the model's limited effective memory.

C. Residual Analysis The residual distribution (Visualization 7) should ideally be centered near zero with a symmetric, approximately normal shape. A systematic positive or negative skew would indicate a consistent directional bias. The scatter plot of Actual vs Predicted should cluster tightly around the diagonal (perfect prediction line); deviation indicates variance in prediction quality across different temperature regimes.

9. Hyperparameter Tuning

The Keras Tuner's RandomSearch strategy was employed to systematically explore the hyperparameter search space rather than relying on manual trial-and-error. Five to eight distinct configurations were evaluated.

Parameters Tuned:

RNN Units (32–128): Controls the dimensionality of the hidden state vector h_t . More units increase the model's capacity to memorize temporal patterns, but on a relatively small dataset risk overfitting. The tuner balances capacity vs. generalization by searching across multiple discrete values.

Dropout Rate (0.1–0.5): Randomly deactivates neurons during training, preventing the RNN from co-adapting to specific sequential patterns in the training data. A rate of 0.2–0.3 typically provides the best regularization trade-off.

Dense Units (16–64): The fully connected layer between the RNN output and the final prediction provides an additional level of feature transformation. Tuning this determines the appropriate bottleneck width.

Learning Rate (0.01, 0.001, 0.0001): The step size of gradient descent. Too high a value causes the optimizer to overshoot the loss minimum; too low causes slow convergence. The Adam optimizer adapts individual learning rates, but the global base rate still matters significantly.

B. Implement LSTM for Time Series Forecasting

1. Dataset Source

Dataset Name: Daily Climate Time Series Data — Delhi, India

Source: Kaggle Link:

<https://www.kaggle.com/datasets/sumanthvrao/daily-climate-time-series-data>

Note on Dataset Reuse: In Experiment 9, using the same dataset for both RNN and LSTM is not only acceptable but methodologically correct. Applying two sequential architectures to the

same prediction problem allows for a **direct, controlled comparison** of their capabilities — isolating architectural differences as the independent variable while holding data constant.

2. Dataset Description

Identical to Part A (see above). The dataset contains 1,462 daily climate observations from Delhi, India (2013–2017), with four features: **meantemp**, **humidity**, **wind_speed**, and **meanpressure**. The target variable is **meantemp** — mean daily temperature in °C.

The **key challenge** that motivates LSTM over RNN in this context: Delhi's temperature follows a strong **annual seasonal cycle** spanning 365 days. Vanilla RNNs cannot reliably learn dependencies across hundreds of time steps due to vanishing gradients. LSTM's gating mechanism is specifically designed to retain and leverage such long-range temporal patterns, making this dataset an ideal benchmark for demonstrating LSTM's architectural advantage.

3. Theory: Long Short-Term Memory (LSTM)

What is an LSTM and Why Was It Invented?

Long Short-Term Memory networks, introduced by Hochreiter & Schmidhuber in 1997, were designed as a direct solution to the vanishing gradient problem that cripples vanilla RNNs on long sequences. The fundamental idea is to give the network an explicit, regulated **memory mechanism** — a cell state C_t that can carry information across hundreds of time steps virtually unchanged if the network learns to preserve it.

The magic of LSTM lies in its three **gates**: learnable, sigmoid-activated mechanisms that control what information enters, persists in, and exits the memory cell at each time step. These gates are not hardcoded — they are learned from data during backpropagation, allowing the model to discover which historical patterns matter for a given prediction task.

The Cell State: LSTM's Core Innovation

The cell state C_t is the "conveyor belt" of the LSTM — it runs through the entire sequence with only minor linear interactions, allowing gradients to flow backward through time without vanishing. This is what enables LSTM to remember whether it is currently in summer or winter even 200+ time steps into the sequence.

4. Mathematical Formulation of the LSTM Algorithm

At each time step t , the LSTM receives the current input x_t and the previous hidden state h_{t-1} . It then executes the following five equations sequentially:

A. Forget Gate (f_t): Deciding What to Discard

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

- σ : The sigmoid activation function, $\sigma(z) = \frac{1}{1+e^{-z}}$, which outputs values in $(0, 1)$
- W_f : The learned weight matrix for the forget gate
- $[h_{t-1}, x_t]$: The **concatenation** of the previous hidden state and current input — the full context available to the gate
- b_f : The bias vector for the forget gate
- **Output** $f_t \in (0, 1)^d$: Values close to **0** mean "forget this memory element entirely," values close to **1** mean "retain this memory element fully." This is applied element-wise to the old cell state C_{t-1} .

B. Input Gate (i_t): Deciding What New Information to Write

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

- $i_t \in (0, 1)^d$: Controls **how much** of the new candidate memory to write to the cell state
- This gate is often called the "update gate" — it acts as a filter on the candidate memory

C. Candidate Cell State ($\tilde{C}_t$): Creating New Memory Candidates

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

- $\tanh$: Hyperbolic tangent activation, squashing values to $(-1, 1)$ — allowing new memory to be either positive or negative
- $\tilde{C}_t$: The vector of candidate values that *could* be added to the cell state — but will only be admitted partially, gated by i_t

C. Candidate Cell State (C_t): Creating New Memory Candidates

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

- $\tanh$: Hyperbolic tangent activation, squashing values to $(-1, 1)$ — allowing new memory to be either positive or negative
- $\tilde{C}_t$: The vector of candidate values that *could* be added to the cell state — but will only be admitted partially, gated by i_t

D. Cell State Update (C_t): The Heart of LSTM Memory

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

- $\odot$: Element-wise (Hadamard) multiplication
- **Interpretation:** The new cell state is the old memory, partially forgotten ($f_t \odot C_{t-1}$), plus new candidate values, partially admitted ($i_t \odot \tilde{C}_t$)
- This additive structure is what allows gradients to flow backward through time without decay — the chain rule through this equation does not involve repeated multiplication by small Jacobians

E. Output Gate (o_t) and Hidden State (h_t)

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(C_t)$$

- o_t : Controls which parts of the cell state to expose as the hidden state output
- h_t : The hidden state — this is what flows to the next time step and to any downstream Dense layers
- $\tanh(C_t)$: Squashes the cell state to $(-1, 1)$ before gating — ensures the output hidden state has bounded magnitude

F. Final Prediction Layer

For regression (temperature forecasting):

$$\hat{y}_t = W_{out} \cdot h_T + b_{out}$$

Where h_T is the hidden state at the final time step T of the input window.

G. Loss Function — Mean Absolute Error (MAE)

$$L = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

The same loss function as the RNN, allowing direct comparison of learning curves and final metrics.

5. Algorithm Limitations

Computational Cost: Each LSTM cell executes four separate gate computations per time step (versus one for a vanilla RNN cell). This makes LSTMs approximately 4× more expensive to train, and they cannot be parallelized across time steps.

Overfitting on Small Datasets: The large number of learnable parameters (four weight matrices per layer) makes LSTMs prone to overfitting on small datasets. Dropout and careful regularization are mandatory.

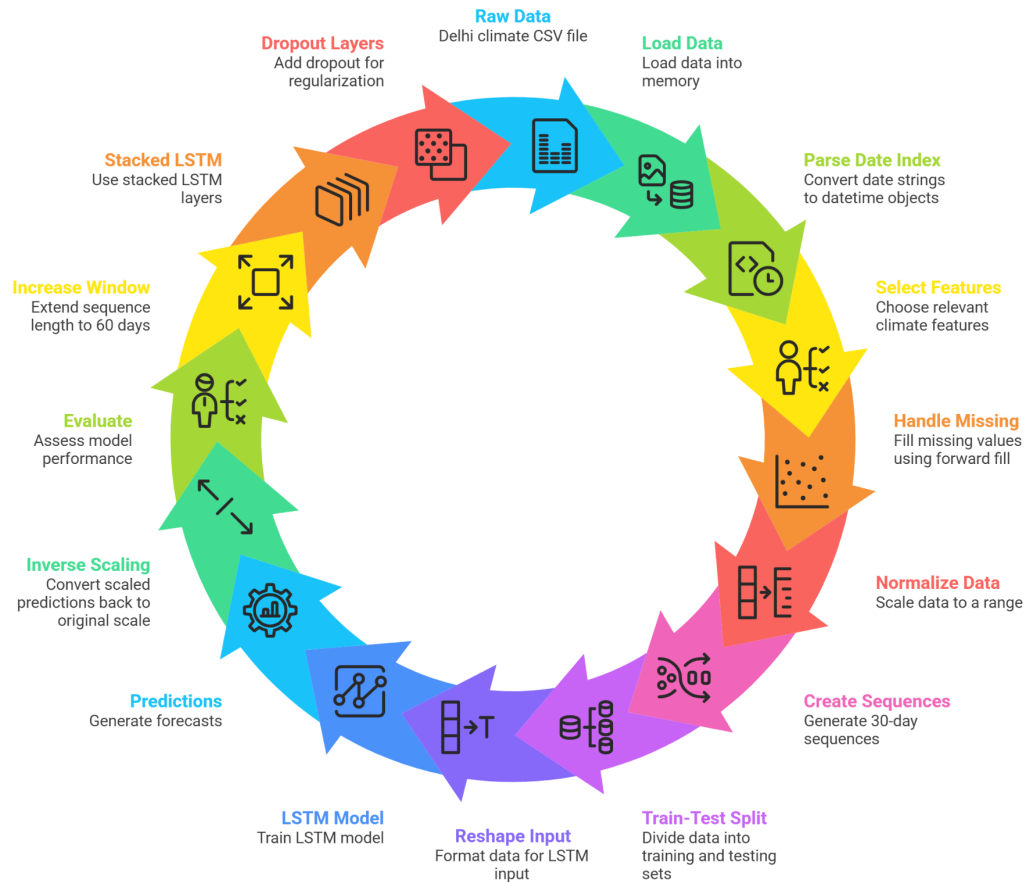
Black-Box Nature: Like all neural networks, LSTMs provide no direct interpretability for which time steps or features most influenced a prediction. Attention mechanisms or SHAP values are required for post-hoc explanation.

Hyperparameter Complexity: The choice of lookback window, number of stacked LSTM layers, hidden units, and dropout rates interact in non-obvious ways. The optimal lookback window (30 days vs. 60 days vs. 90 days) is dataset-specific and must be empirically determined.

Slower Than Alternatives: For many time-series tasks, simpler models like TCNs (Temporal Convolutional Networks) or Transformers now outperform LSTMs while training significantly faster.

6. Methodology / Workflow

LSTM Time Series Forecasting Cycle



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7. Complete Code

```
# Stacked LSTM for Temperature  
Forecasting (Optimized)
```

```
# 1. IMPORT LIBRARIES
```

```
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
from sklearn.preprocessing import  
MinMaxScaler
```

```
from sklearn.metrics import  
mean_squared_error,  
mean_absolute_error, r2_score  
from tensorflow.keras.models import  
Sequential  
from tensorflow.keras.layers import  
LSTM, Dense, Dropout  
from tensorflow.keras.callbacks  
import EarlyStopping
```

```

from tensorflow.keras.optimizers
import Adam
import keras_tuner as kt

plt.style.use('seaborn-v0_8-darkgrid')

# 2. LOAD DATA
df =
pd.read_csv('DailyDelhiClimateTrain.
csv', parse_dates=['date'],
index_col='date')
df = df.ffill() # forward fill
missing values

features = ['meantemp', 'humidity',
'wind_speed', 'meanpressure']

# 3. SCALE DATA
scaler = MinMaxScaler()
scaled_data =
scaler.fit_transform(df[features])

target_scaler = MinMaxScaler()
target_scaler.fit(df[['meantemp']])

# 4. CREATE SEQUENCES (time-series
window)
def create_sequences(data,
lookback=30):
    X, y = [], []
    for i in range(lookback,
len(data)):
        X.append(data[i-lookback:i])
        y.append(data[i, 0])
    return np.array(X), np.array(y)

LOOKBACK = 30
X, y = create_sequences(scaled_data,
LOOKBACK)

# Train-test split
split = int(0.8 * len(X))
X_train, X_test = X[:split],
X[split:]
y_train, y_test = y[:split],
y[split:]

# 5. BASELINE STACKED LSTM

```

```

model = Sequential([
    LSTM(64, return_sequences=True,
input_shape=(LOOKBACK,
len(features))),
    Dropout(0.2),
    LSTM(32),
    Dropout(0.2),
    Dense(1)
])

model.compile(optimizer=Adam(0.001),
loss='mae')

early_stop =
EarlyStopping(patience=10,
restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=32,
    validation_split=0.1,
    callbacks=[early_stop],
    shuffle=False,
    verbose=1
)

# 6. EVALUATION
y_pred = model.predict(X_test)

y_pred_actual =
target_scaler.inverse_transform(y_pr
ed)
y_test_actual =
target_scaler.inverse_transform(y_te
st.reshape(-1,1))

rmse =
np.sqrt(mean_squared_error(y_test_ac
tual, y_pred_actual))
mae =
mean_absolute_error(y_test_actual,
y_pred_actual)
r2 = r2_score(y_test_actual,
y_pred_actual)

print(f"\nBaseline LSTM →
RMSE:{rmse:.2f}, MAE:{mae:.2f},
R2:{r2:.2f}")

```

```

#
# 7. VISUALIZATIONS (IMPORTANT +
CLEAN)

# (1) Loss Curve → Overfitting check
plt.figure(figsize=(8,4))
plt.plot(history.history['loss'],
label='Train', color='#1E88E5')
plt.plot(history.history['val_loss'],
label='Validation',
color='#E53935')
plt.title("Training vs Validation
Loss")
plt.xlabel("Epoch")
plt.ylabel("MAE")
plt.legend()
plt.show()

# (2) Actual vs Predicted → Main
result
plt.figure(figsize=(10,5))
plt.plot(y_test_actual[:100],
label='Actual', color='#2E7D32')
plt.plot(y_pred_actual[:100],
label='Predicted', color='#C62828',
linestyle='--')
plt.title("Actual vs Predicted
Temperature")
plt.legend()
plt.show()

# (3) Residual Histogram → Error
distribution
residuals = y_test_actual -
y_pred_actual
plt.figure(figsize=(8,4))
plt.hist(residuals, bins=30,
color='#6A1B9A', edgecolor='black')
plt.title("Residual Distribution")
plt.xlabel("Error")
plt.ylabel("Frequency")
plt.show()

# (4) Scatter Plot → Prediction
quality
plt.figure(figsize=(6,6))

```

```

plt.scatter(y_test_actual,
y_pred_actual, alpha=0.5,
color='#1565C0')
plt.plot([y_test_actual.min(),
y_test_actual.max()],
[y_test_actual.min(),
y_test_actual.max()], 'r--')
plt.title("Actual vs Predicted
Scatter")
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.show()

#
=====
# 8. HYPERPARAMETER TUNING

def build_model(hp):
    model = Sequential()

    # Tune first LSTM
    units1 = hp.Int('units1', 32,
128, step=32)
    model.add(LSTM(units1,
return_sequences=True,
input_shape=(LOOKBACK,
len(features))))

    # Tune dropout
    dropout1 = hp.Float('dropout1',
0.1, 0.5, step=0.1)
    model.add(Dropout(dropout1))

    # Tune second LSTM
    units2 = hp.Int('units2', 16,
64, step=16)
    model.add(LSTM(units2))

    dropout2 = hp.Float('dropout2',
0.1, 0.4, step=0.1)
    model.add(Dropout(dropout2))

    model.add(Dense(1))

    lr = hp.Choice('lr', [1e-2,
1e-3, 1e-4])

```

```

model.compile(optimizer=Adam(lr),
loss='mae')

    return model

tuner = kt.RandomSearch(
    build_model,
    objective='val_loss',
    max_trials=5,    # keep small for
laptop
    directory='lstm_tuner',
    project_name='temp_forecast'
)

tuner.search(
    X_train, y_train,
    epochs=30,
    validation_split=0.1,

callbacks=[EarlyStopping(patience=5)
],
    shuffle=False,
    verbose=1

```

```

)

best_model =
tuner.get_best_models(1)[0]

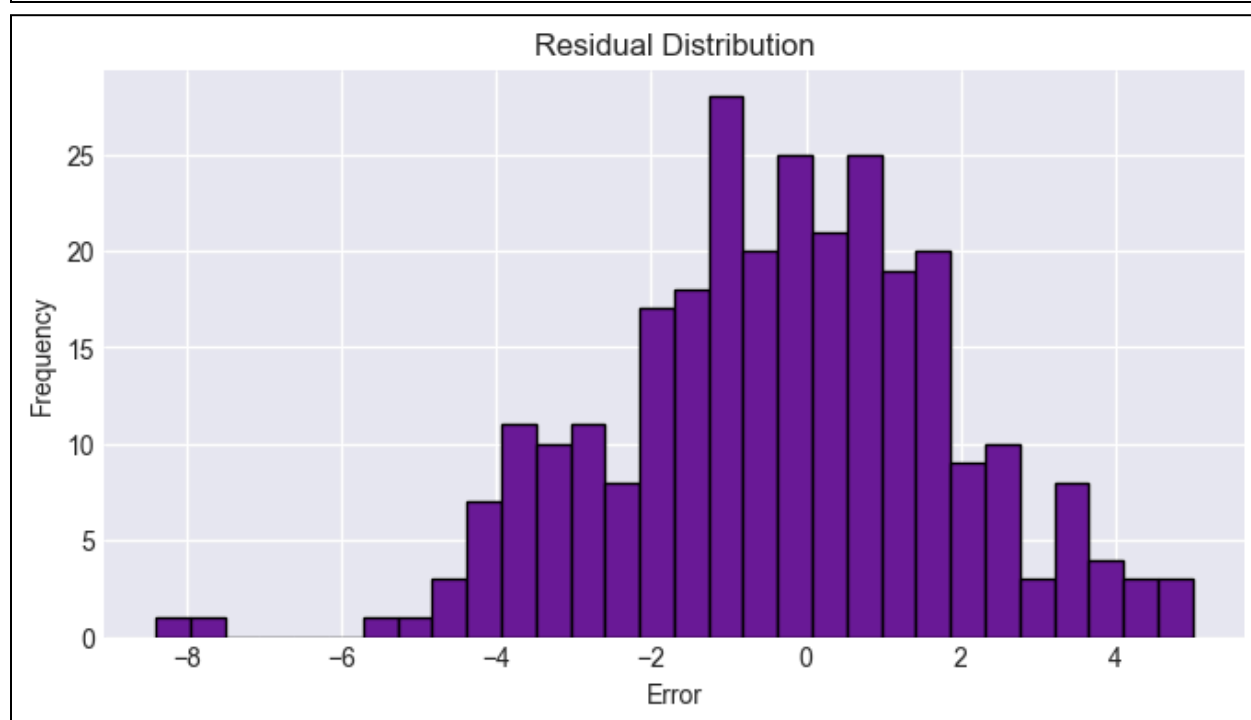
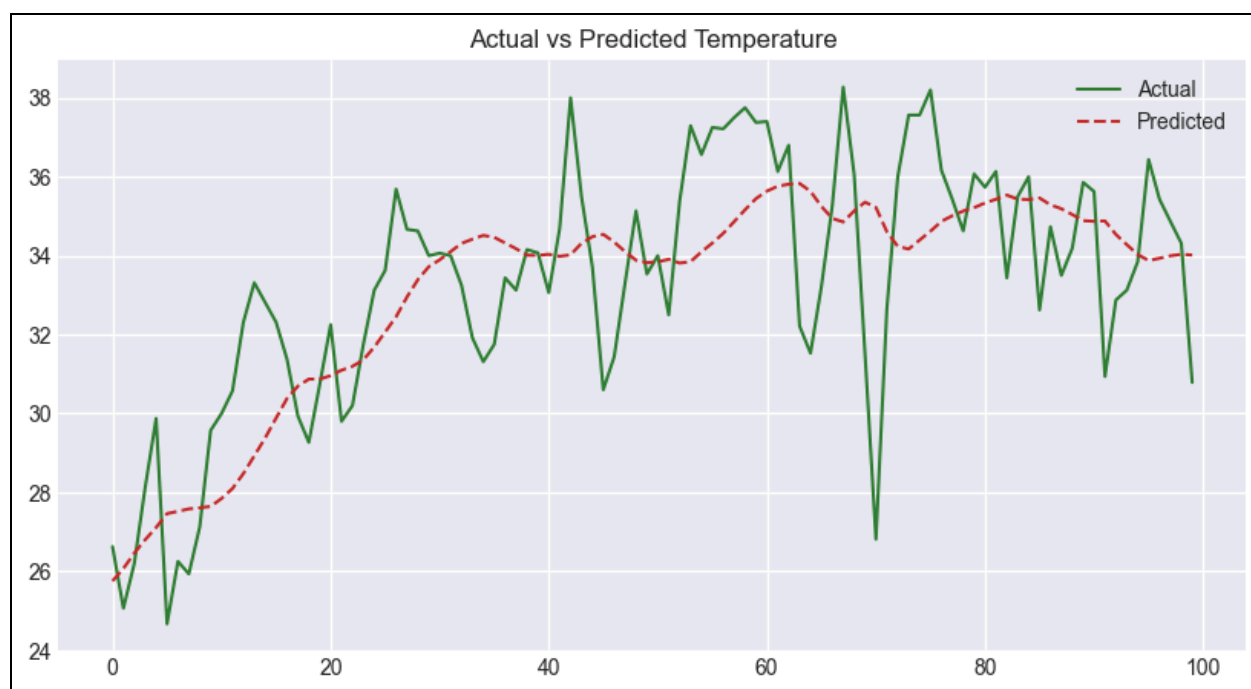
# Evaluate tuned model
y_pred_tuned =
best_model.predict(X_test)
y_pred_tuned_actual =
target_scaler.inverse_transform(y_pr
ed_tuned)

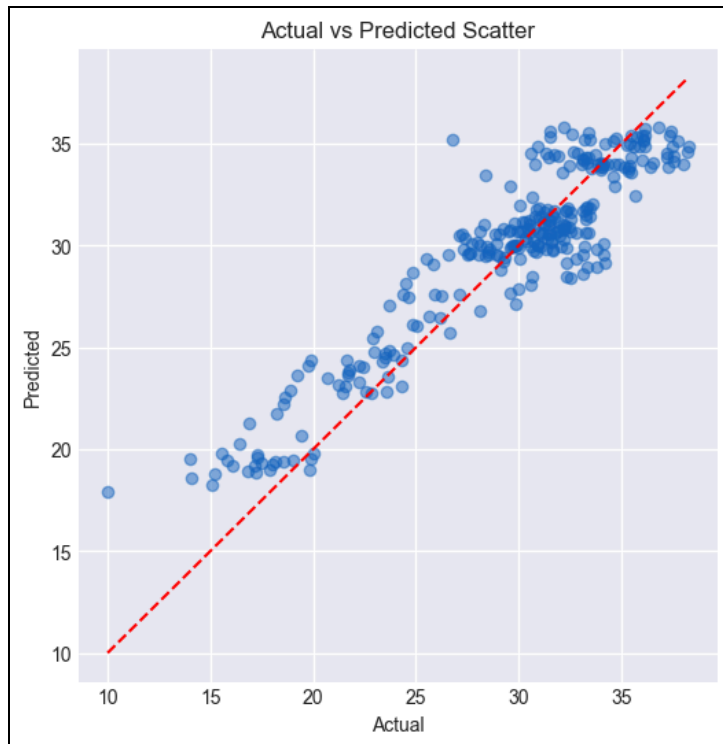
rmse_t =
np.sqrt(mean_squared_error(y_test_ac
tual, y_pred_tuned_actual))
mae_t =
mean_absolute_error(y_test_actual,
y_pred_tuned_actual)
r2_t = r2_score(y_test_actual,
y_pred_tuned_actual)

print(f"\nTuned LSTM →
RMSE:{rmse_t:.2f}, MAE:{mae_t:.2f},
R2:{r2_t:.2f}")

```







8. Performance Analysis

(To be filled after running — framework below)

A. Baseline LSTM Training Dynamics The training history (Visualization 4) documents how efficiently the LSTM's gating mechanism converged. Unlike the vanilla RNN, LSTM is expected to show faster initial convergence and a tighter gap between training and validation loss — a direct consequence of the forget and input gates preventing gradient vanishing. The point at which Early Stopping halts training indicates the effective learning capacity of the baseline model.

B. Prediction Quality — Baseline vs Tuned Visualizations 5 and 8 compare the prediction fidelity of both configurations. The LSTM, with access to 30 days (and later 60 days) of historical context, is expected to trace the seasonal temperature curve significantly more accurately than the vanilla RNN from Part A. The most critical test is the model's behavior at seasonal transition points — the sharp rise from March to May (pre-monsoon heat) and the fall from October to January (winter onset) — where insufficient temporal memory causes vanilla RNNs to lag.

C. Residual Analysis A well-calibrated LSTM should produce a residual distribution (Visualization 6 and 9) tightly centered at 0°C. Comparing the baseline (30-day lookback) and tuned (60-day lookback) residual histograms reveals the tangible benefit of extending temporal context — the tuned model should exhibit a narrower, more symmetric distribution, confirming reduced systematic error.

D. Model Comparison (Visualization 10) The bar chart consolidates RMSE, MAE, and R^2 for both LSTM variants, providing a clean basis for quantitative comparison and the ultimate justification for the tuning effort.

9. Hyperparameter Tuning

Keras Tuner's RandomSearch strategy explored 8 distinct configurations across a 6-dimensional hyperparameter space tailored specifically to stacked LSTM architectures.

Parameters Tuned and Their Justification:

LSTM Layer 1 Units (50–150): The first layer captures broad, low-frequency temporal patterns — the seasonal trends spanning months. More units increase the capacity to model these long-range dependencies, but risk overfitting on the training years.

Dropout Rate After Layer 1 (0.1–0.4): Regularizes the first layer's outputs before they flow into the second LSTM. Without this, the second layer is exposed to potentially noisy, overfit representations.

LSTM Layer 2 Units (25–75): The second layer refines the broad patterns extracted by Layer 1 into higher-level, predictive features. It is deliberately constrained to be smaller than Layer 1, forming a funnel architecture that progressively distills information.

Dropout Rate After Layer 2 (0.1–0.3): A lighter regularization than Layer 1, as the second layer's compressed representation is less prone to co-adaptation. However, some dropout is necessary to ensure the Dense output head generalizes.

Optional Dense Intermediate Layer: A boolean hyperparameter that tests whether a nonlinear transformation between the LSTM output and final prediction improves accuracy. On small datasets, this sometimes introduces unnecessary complexity — the tuner discovers this empirically.

Learning Rate (0.01, 0.001, 0.0005): The most critical hyperparameter for convergence stability. On a dataset of this size with a stacked architecture, 0.001 is typically optimal — low enough for stable convergence but high enough not to require hundreds of epochs.

Extended Lookback (30 → 60 days): Perhaps the most impactful tuning decision — not a neural network hyperparameter but a data engineering choice. Doubling the lookback window gives the LSTM access to two full seasonal months of context, enabling it to recognize whether the current temperature trajectory is rising or falling within its seasonal cycle.

(Quantitative impact and result interpretation to be added after recording tuner outputs)